

Lecture 17

STAT 109: Introductory Biostatistics

Lecture 17: Balancing Type I and Type II Errors in Hypothesis Tests

What is the probability of making a mistake in a test?

Learning outcomes

- Define Type I and Type II errors in terms of the null and alternative hypotheses in a test.
 - Describe the result of either a Type I or Type II error in the context of a specific research question.
 - Define power both in terms of the null and alternative hypotheses in a test and as the opposite of the probability of making a Type II error.
 - Balance the two types of error in a test by:
 - using the consequence of a Type I error to inform the choice of the level of significance (e.g. how small the p-value needs to be to reject the null hypothesis), and
 - increasing the sample size until a reasonable value for power / probability of a Type II error is achieved (assuming a particular alternative hypothesis is true).
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The trial-by-jury analogy

Hypothesis testing works like a **trial by jury**:

- **Null hypothesis** (H_0) = the default assumption of **innocence**. We assume “no effect” or “no difference” until there is strong evidence otherwise.
 - **P-value** = the **weight of the evidence** against the null. A small p-value means the data would be unlikely if H_0 were true — strong evidence against “innocence.” A large p-value means the data are plausible under H_0 — we do not have enough evidence to reject the default.
 - **Reject** H_0 = like a guilty verdict: we conclude the evidence is strong enough to reject the default (innocence).
 - **Fail to reject** H_0 = like a not-guilty verdict: we do not have enough evidence to reject the default; we are not saying H_0 is “true,” only that we cannot reject it.
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Performing a hypothesis test (recap)

The usual flow:

1. **State the hypotheses:** H_0 (null, default) and H_a (alternative).
2. **Collect data** and compute a test statistic (e.g. sample proportion, count).
3. **Compute the p-value:** the probability of observing data as or more extreme than what we saw, **if H_0 were true**.
4. **Decide:** If the p-value is small (e.g. below a chosen significance level α), **reject** H_0 ; otherwise **fail to reject** H_0 .

Every decision is made in the face of uncertainty: H_0 might really be true or false, and we might reject it or not. So we can make **mistaken decisions**.

Decision vs. reality

Our **decision** (reject or fail to reject H_0) can be right or wrong depending on **reality** (whether H_0 is actually true or false):

	Reality: H_0 true	Reality: H_0 false
Decision: Reject H_0	Type I error (false alarm)	Correct — we reject when we should (power)
Decision: Fail to reject H_0	Correct — we keep the default when it's true	Type II error (missed opportunity)

- **Type I error:** Rejecting H_0 when H_0 is actually true.
- **Type II error:** Failing to reject H_0 when H_0 is actually false.
- **Power:** Probability of correctly rejecting H_0 when H_0 is false. So **Power** = $1 - \text{P}(\text{Type II error})$.

Thinking about mistaken decisions

- **Type I error** is a “false alarm”: we conclude there is an effect (reject the null) when there isn't. In the trial analogy, we convict an innocent person.
- **Type II error** is a “missed opportunity”: we fail to conclude there is an effect when there really is one. In the trial analogy, we fail to convict someone who is guilty.

We cannot make both error probabilities arbitrarily small at once with a fixed sample size. So we **balance** them: choose a significance level α (max acceptable Type I error) and then try to achieve reasonable power (low Type II error) via sample size or design.

Defining Type I error, Type II error, and power

- **Type I error:** Rejecting the null hypothesis when it is actually true. We control this with the **significance level** α (e.g. $\alpha = 0.05$): we reject H_0 only when the p-value $< \alpha$.
- **Type II error:** Failing to reject the null hypothesis when it is actually false. Its probability depends on the true state of nature (e.g. how far the true p is from the null value), the sample size, and α .
- **Power:** The probability of correctly rejecting H_0 when H_0 is false. **Power** = $1 - \text{P}(\text{Type II error})$. Higher power means we are more likely to detect a real effect when it exists.

Balancing the two types of error

- **When Type I error is more serious** (e.g. approving a harmful drug): use a **smaller** α (e.g. 0.01) so we require stronger evidence (smaller p-value) before rejecting H_0 .
 - **When Type II error is more serious** (e.g. missing a beneficial treatment): we want **higher power**. We can increase power by increasing the sample size or, if acceptable, using a somewhat larger α (e.g. 0.10).
 - **Increasing sample size** generally increases power (reduces Type II error) without changing the Type I error rate α .
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Example 1: The “lucky” coin

Setup: We want to know if a coin lands on heads more often than not (a “lucky” coin for heads).

State the hypotheses

- H_0 : The coin is fair; $p = 0.5$ (probability of heads).
- H_a : The coin is lucky for heads; $p > 0.5$.

Define errors in context

- **Type I error:** Concluding that the coin is “lucky” when it actually is not (false alarm).
- **Type II error:** Failing to conclude that the coin is “lucky” when it actually is (missed opportunity).

Define power

- Power = probability of correctly concluding the coin is lucky when it really is ($p > 0.5$); i.e. Power = $1 - P(\text{Type II error})$.

Balance the errors

- Choose α (e.g. 0.05) to control how often we falsely declare the coin lucky. To reduce the chance of missing a truly lucky coin, increase the number of flips (sample size) to increase power.
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Example 2: The “lucky” die

Setup: We want to know if a die lands on a particular side (e.g. 6) more often than it should.

State the hypotheses

- H_0 : The die is fair; $p = 1/6$ (probability of that side).
- H_a : The die is “lucky” for that side; $p > 1/6$.

Define errors in context

- **Type I error:** Concluding that the die is “lucky” when it really isn’t.
- **Type II error:** Failing to conclude that the die is “lucky” when it really is.

Define power

- Power = probability of correctly concluding the die is lucky when it really is; Power = $1 - P(\text{Type II error})$.

Balance the errors

- Choose α to limit false claims of a lucky die. Increase the number of rolls to improve power if we care about not missing a biased die.
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Example 3: Medical treatment effectiveness

Setup: A new medical treatment is studied. We want to know if it is effective in **more than 10%** of cases.

State the hypotheses

- H_0 : The treatment is effective in exactly 10% of cases; $p = 0.10$
- H_a : The treatment is effective in more than 10% of cases; $p > 0.10$.

Define errors in context

- **Type I error:** Concluding that the treatment is effective in more than 10% of cases when it actually isn't (false discovery — we might adopt an ineffective treatment).
- **Type II error:** Failing to conclude that the treatment is effective when it actually helps more than 10% of patients (missed opportunity — we might abandon a beneficial treatment).

Define power

- Power = probability of correctly concluding the treatment is effective when it really is (e.g. when true $p > 0.10$);
Power = $1 - P(\text{Type II error})$.

Balance the errors

- If falsely approving an ineffective treatment is very serious, use a **smaller** α (e.g. 0.01). If missing a beneficial treatment is very serious, design the study with **larger sample size** (and possibly a somewhat larger α) to achieve higher power.

Increasing power

- Increase the sample size; optionally use a slightly larger α if Type II error is the main concern; or focus the alternative (e.g. a specific $p > 0.10$) to design for that effect size.

Summary of terms

Term	Meaning
Null hypothesis (H_0)	The null model; a statement of no effect or no difference; the default assumption (e.g. “innocence”).
Alternative hypothesis (H_a)	The statement we consider if we reject H_0 (e.g. there is an effect or a difference).
P-value	The probability of observing data as or more extreme than what we saw, if the null model were true. Small p-value = strong evidence against H_0 ; large p-value = we cannot confidently reject H_0 .
Type I error	Rejecting the null hypothesis when it is actually true (false alarm).
Type II error	Failing to reject the null hypothesis when it is actually false (missed opportunity).
Power	Probability of correctly rejecting H_0 when H_0 is false; Power = $1 - P(\text{Type II error})$.

Use the **trial-by-jury** analogy: H_0 = default assumption of innocence, p-value = weight of evidence, and the Decision vs. Reality table to see how Type I, Type II, and power fit together.